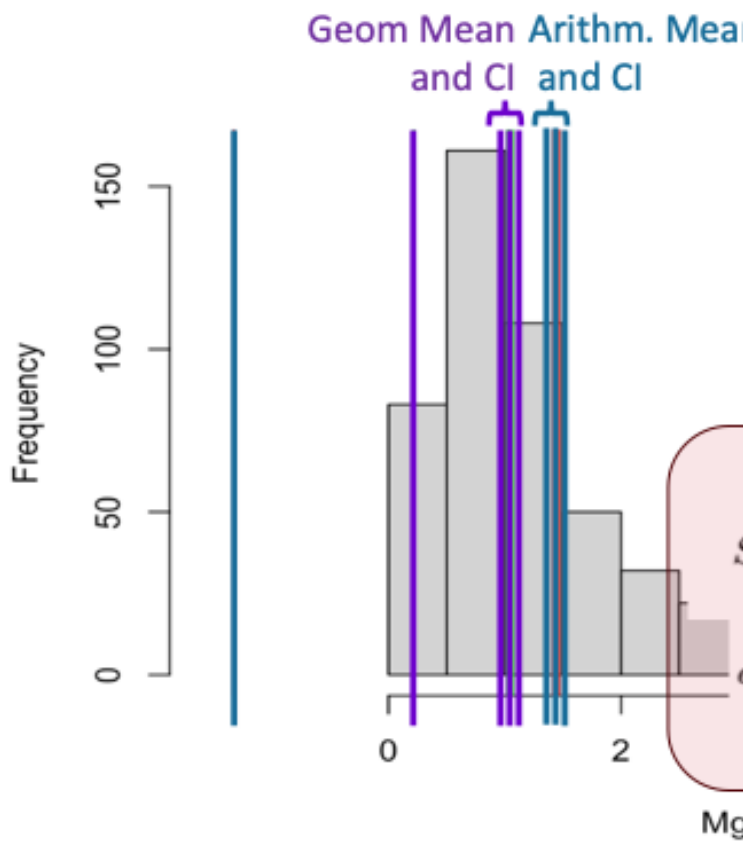
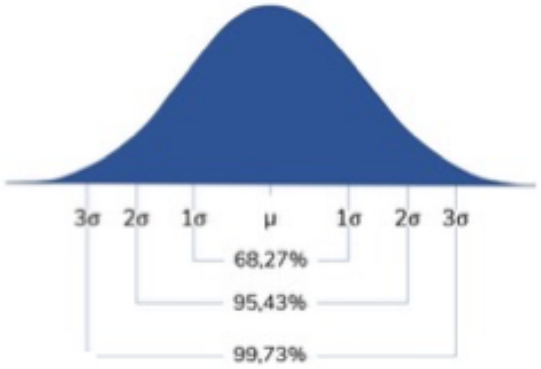




Aitchison simplex with Aitchison metric

$$S^D = \{\mathbf{x} = (x_1, \ldots, x_D) \mid x_i > 0, \sum x_i = 1\}$$

$$d_A(\mathbf{x}, \mathbf{y}) = \sqrt{\frac{1}{2D} \sum_{i=1}^{D} \sum_{j=1}^{D} [\ln(x_i/x_j) - \ln(y_i/y_j)]^2}$$

TECHNISCHE UNIVERSITÄT BERLIN

Time Series Analysis – parts III and IV

Dr. Pedro Alencar | FG AI and Land Use Change| NFDI4earth | March 17, 2026

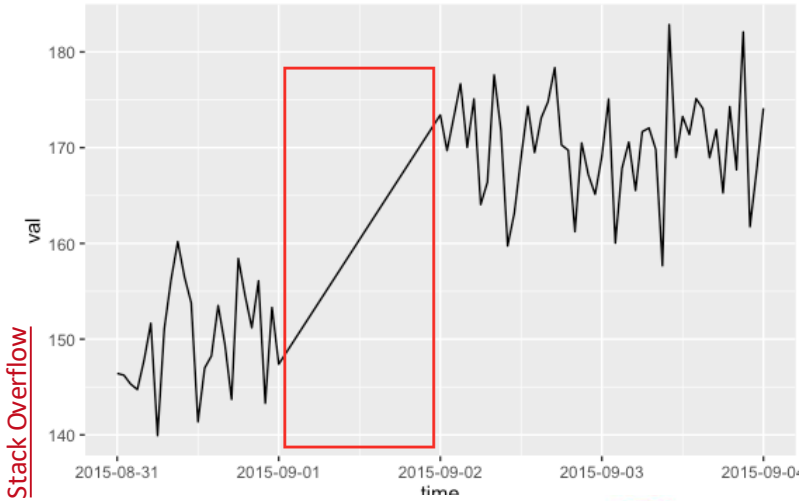
Time series analysis – Why?

- You may have noticed that many interesting processes happen in time
  - Climate
  - Water
  - Markets
  - Politics
  - Mass extinctions...

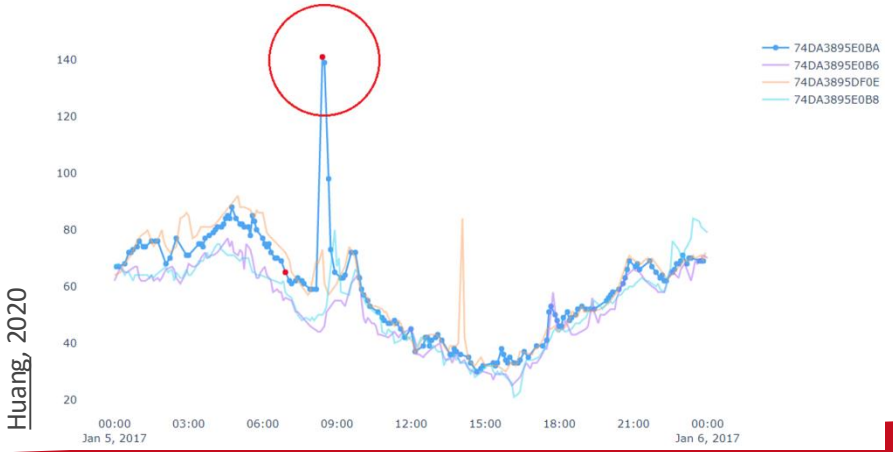
Time series analysis – Why?

- Time series does not work just like other data
  - Observations are not independent (today's temperature depends on yesterday's).
  - Standard tools (e.g., t-tests, linear regression) can give misleading results as temporal dependence (autocorrelation) is ignored.

Time series analysis – Why?



Stack Overflow



Huang, 2020

- The real-world data is messy
  - Outliers, missing values, non-normal distributions are common.
- The future is messier
  - And we want to forecast it

Review of Statistical Fundamentals

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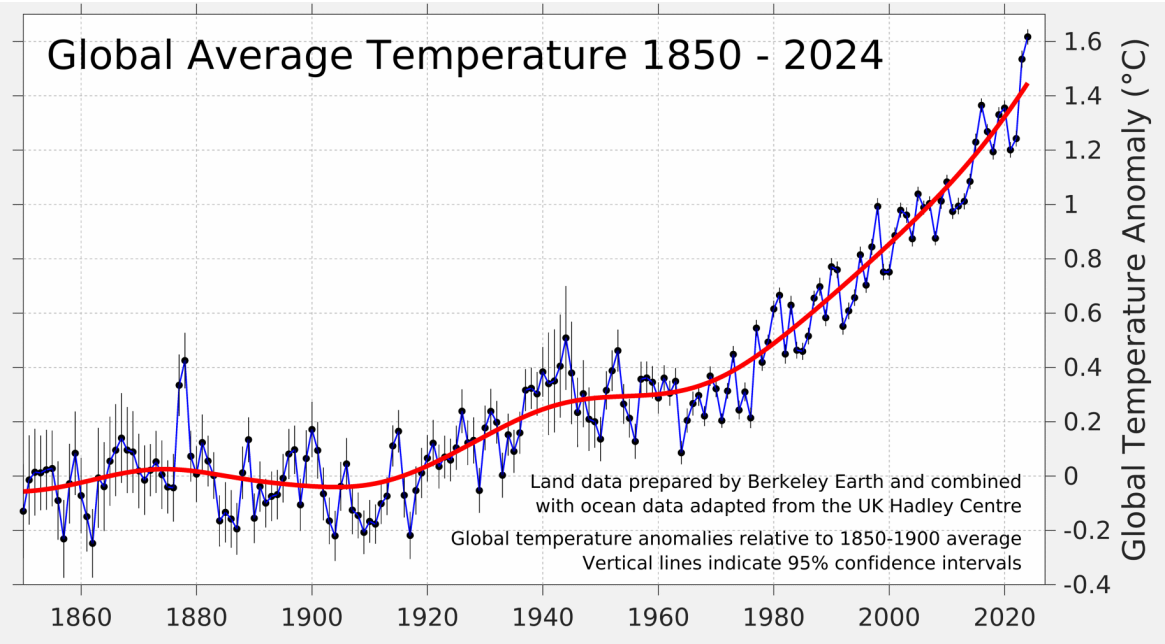
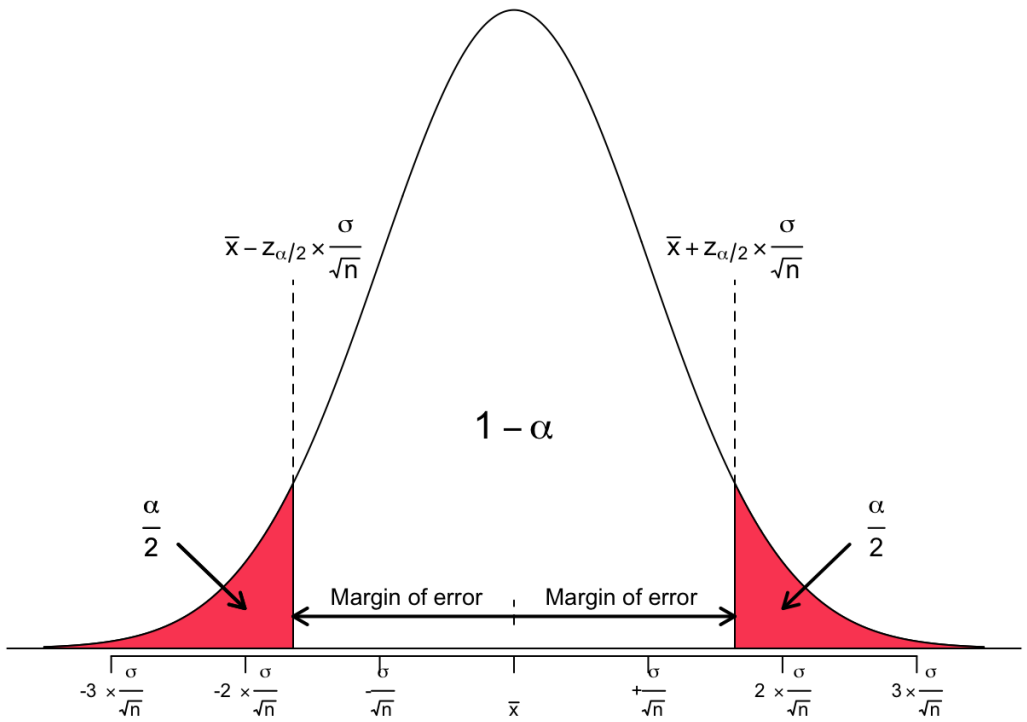
What is a time series?



Rohde, 2025

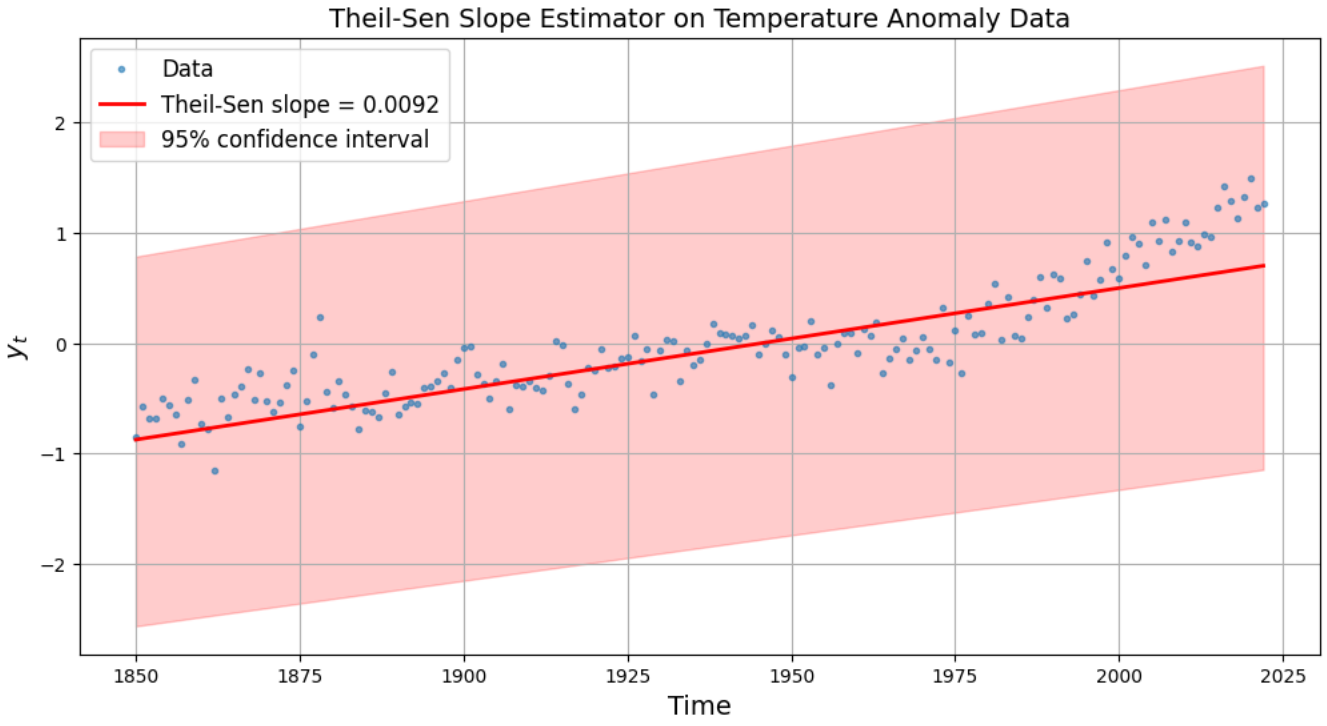
- A sequence of data points collected over time (temperature, precipitation, stock prices…)
- Key characteristic: observations are ordered and **often** dependent on previous values
- Key notation:
  - $y_1, y_2, \ldots, y_n$ observations
  - at times $t_1, t_2, \ldots, t_n$

Confidence Intervals Review



- A confidence interval gives a **range** of plausible values for an unknown parameter
- A 95% CI means: if we repeated the experiment many times, **95% of the intervals would contain the true value**
- General form: Estimate ± (Critical Value × Standard Error)

Confidence Intervals Review

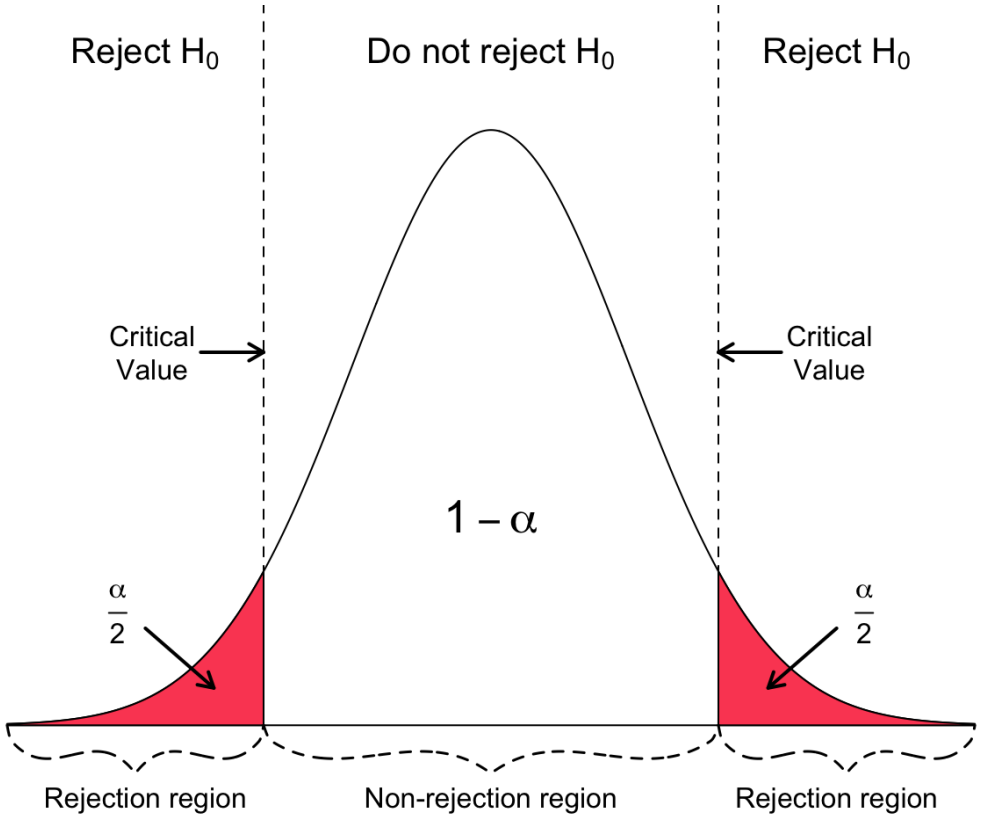


- Formula:
  - $CI = \bar{x} \pm z_{\alpha/2} \cdot (\sigma / \sqrt{n})$

Confidence level (e.g., 95%) ≠ probability that the true value is in the interval

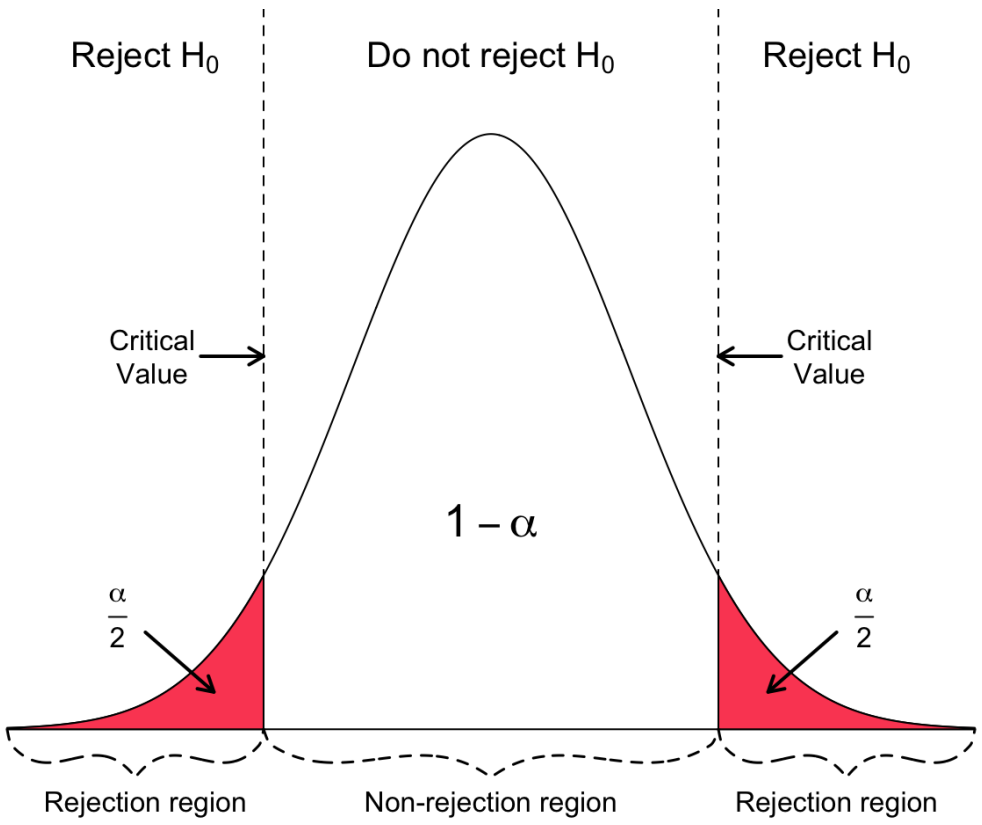
***"interval estimation** gives a probabilistic statement, relating the given interval to the probability that this interval actually contains the true (unknown) population parameter."*

Hypothesis Testing Review



- As statisticians/data-analysts, we lack knowledge of the population, knowing only part of it (sample)
- With HT we can provide “statistical inferences about populations based on samples taken from the population.”
- For that, we elaborate two hypothesis:
  - H_0 – the null hypothesis
  - H_1 – the alternative hypothesis

Hypothesis Testing Review



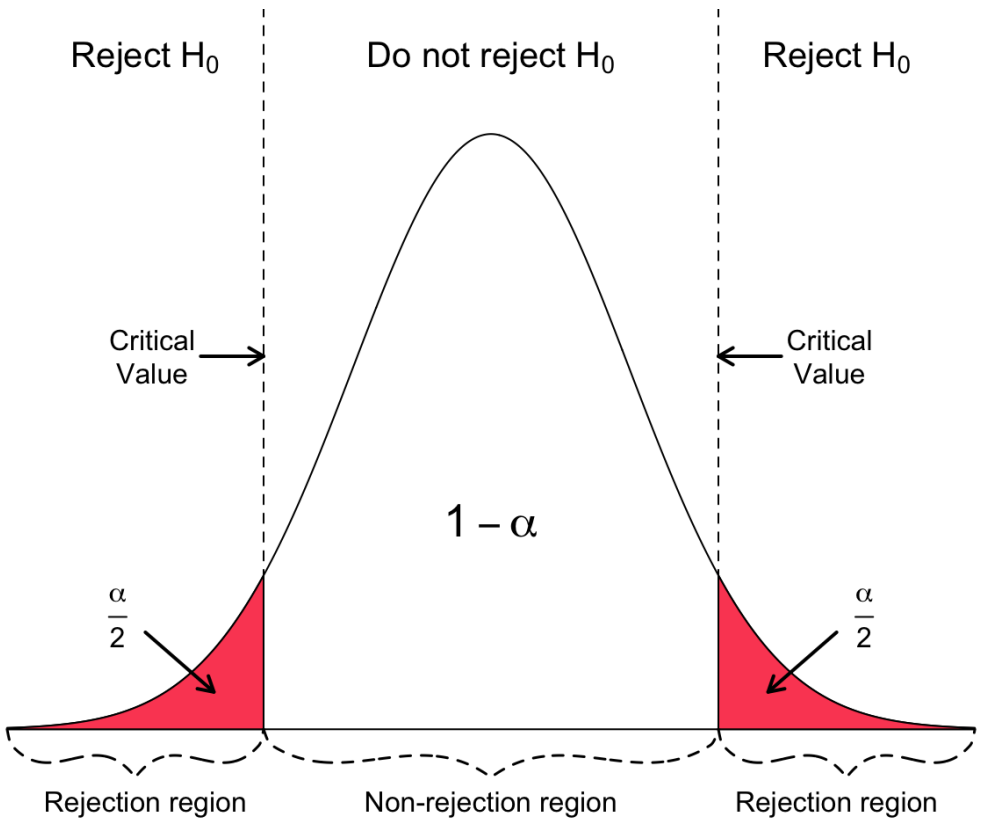
- The null hypothesis for a hypothesis test concerning a population mean, μ, is expressed:

$$H_0: \mu = \mu_0,$$

- The alternative hypothesis is expressed as

$$H_1: \mu \neq \mu_0.$$

Hypothesis Testing Review



- H_0 (Null Hypothesis): No effect / no trend / no difference
- H_1 (Alternative Hypothesis): There is an effect / trend / difference
- p-value: Probability of observing the data (or more extreme) if H_0 is true
- Significance level α: Threshold for rejecting H_0 (commonly 0.05)

If p-value ≤ α → Reject H_0

If p-value > α → Fail to reject H_0

Autocorrelation

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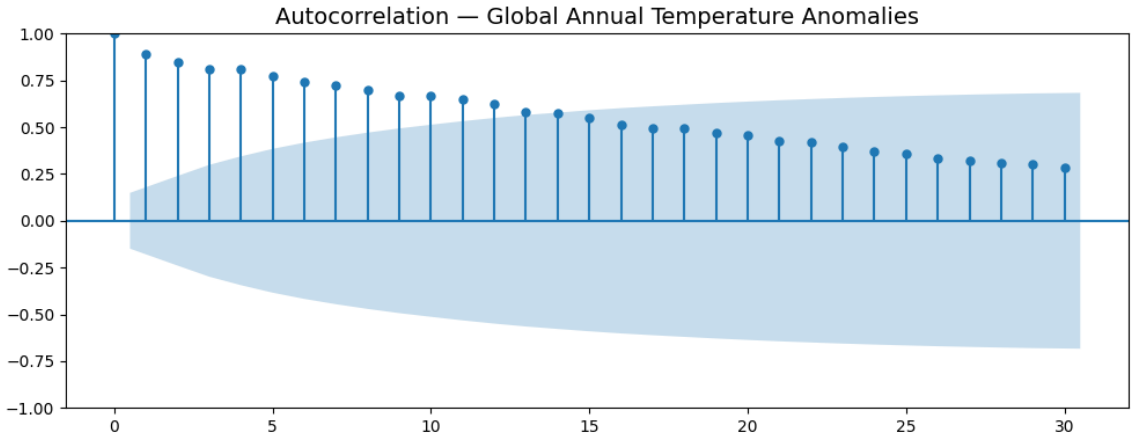
What is autocorrelation?



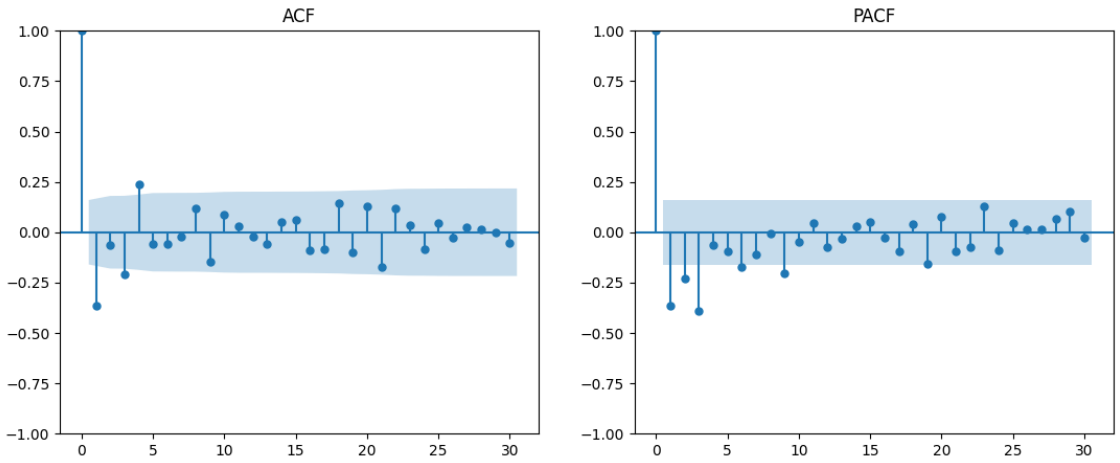
- Autocorrelation is a measure of the internal correlation of a time series.
- It is a representation of the degree of similarity between the time series and a lagged version of itself.

Formula:

$$\hat{\rho}(k) = \frac{\hat{\gamma}(k)}{\hat{\gamma}(0)} = \frac{\sum_{t=1}^{n-k}(y_{t+k} - \bar{y})(y_t - \bar{y})}{\sum_{t=1}^{n}(y_t - \bar{y})^2}$$

Note that $\bar{y} = \frac{1}{n}\sum_{t=1}^{n} y_t$, is constant

What is autocorrelation?

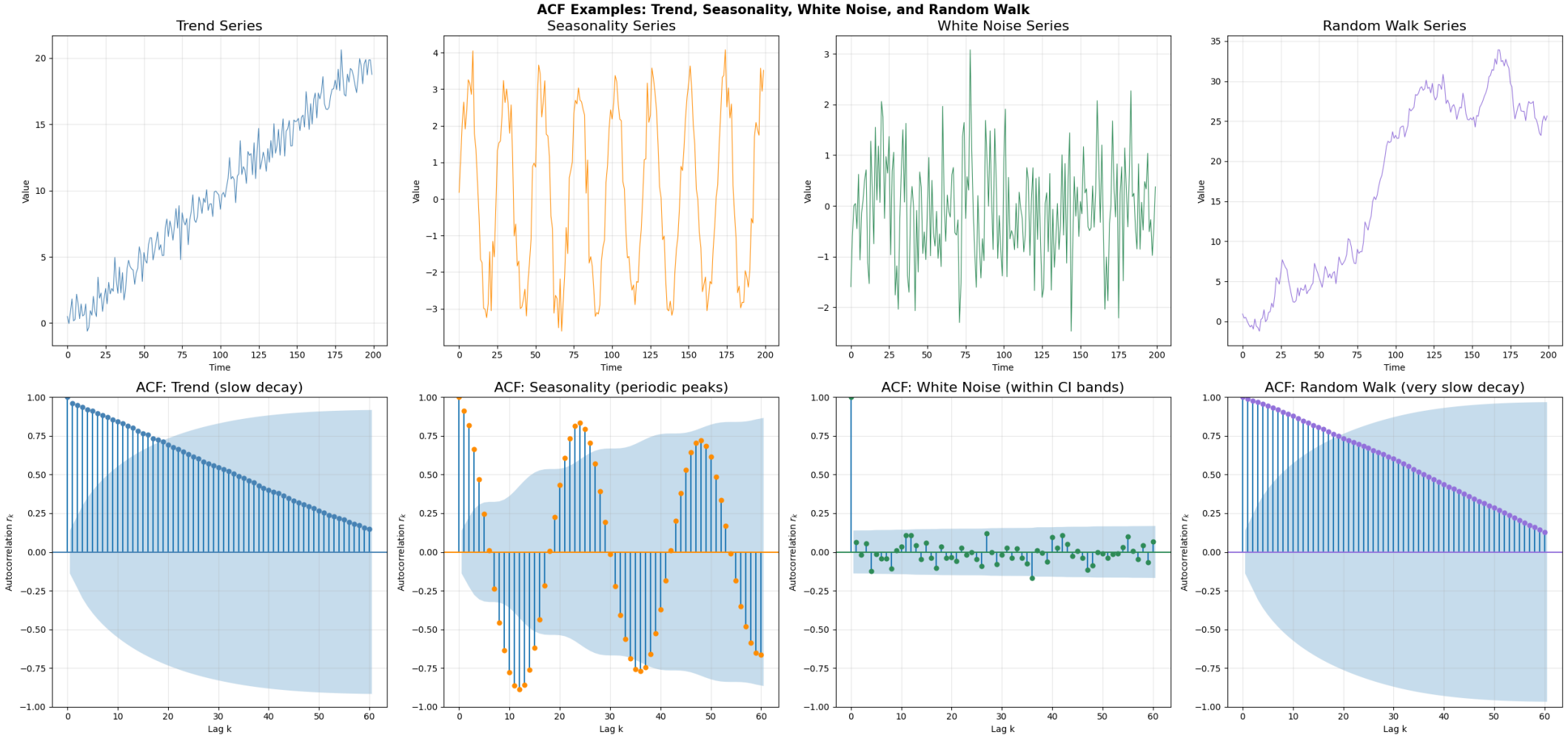


- Similarly, *Partial* Autocorrelation is a measure of internal correlation after removing the linear influence of all intermediate lags

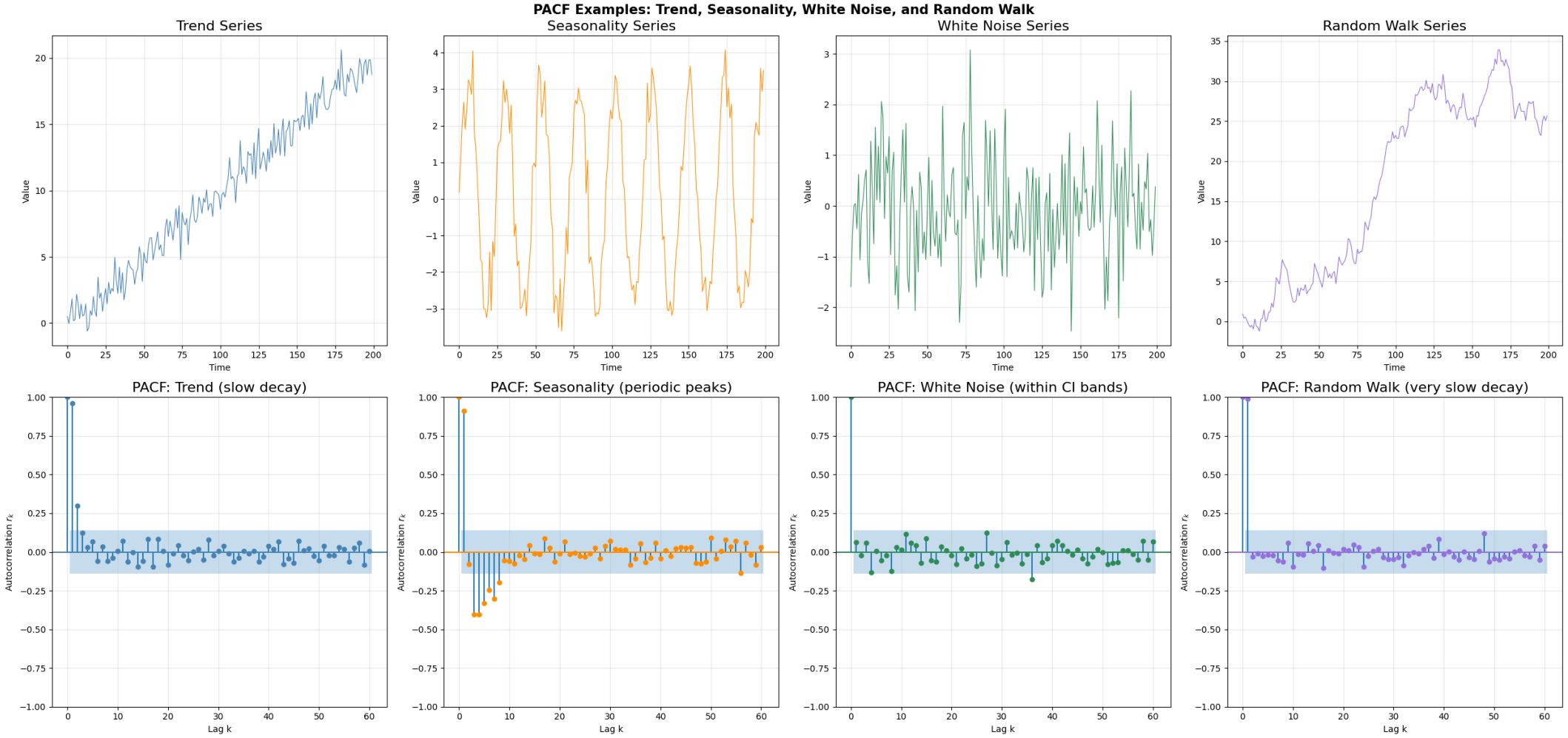
- $\rho_{kk} = \mathrm{Corr}(y_t - \hat{y}t,\ yt - k - \widehat{y_{t-k}})$

ACF and PACF are important to model times series.

Autocorrelation Function (ACF)



Partial Autocorrelation Function (PACF)



Trends Analysis – Mann Kendall test

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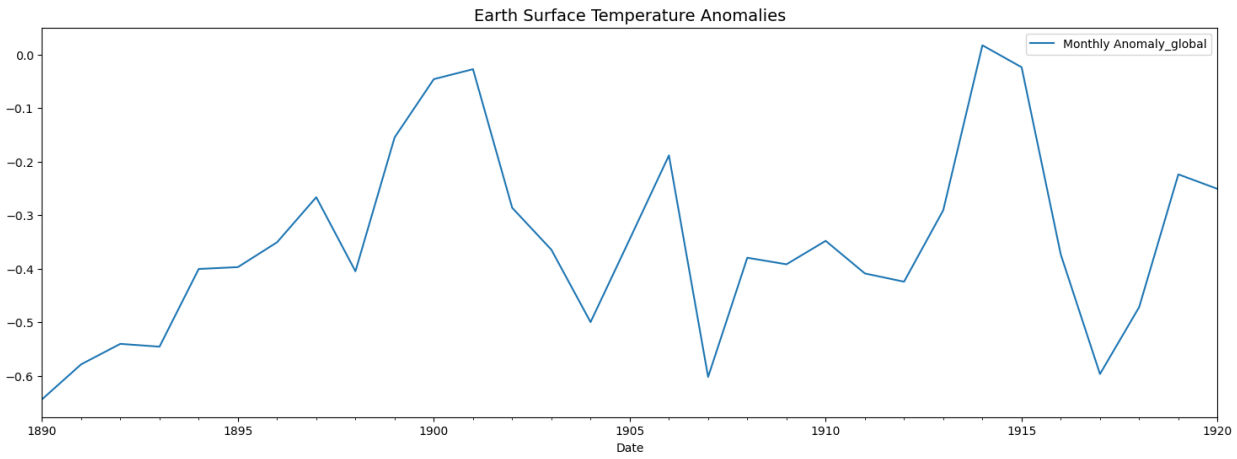
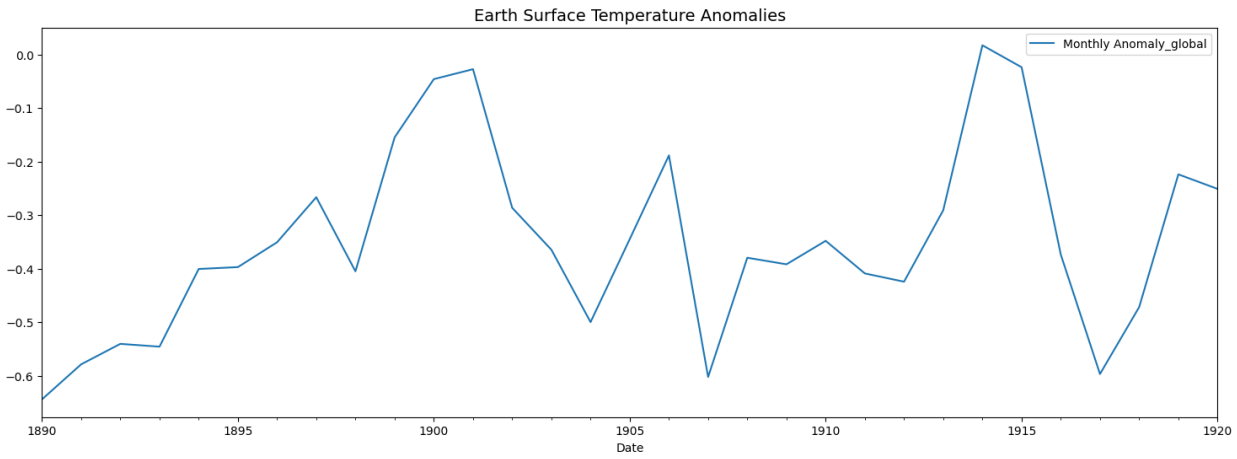
Detecting Trends in Time Series



Question: is there a <u>systemic</u> increase or decrease over time?

Mann-Kendall Test



- Non-parametric test for monotonic trend (upward or downward) - No assumptions about data distribution - Based on comparing all pairs of observations

- **How it works:**

1. For each pair (y_j, y_k) where j > k, compute sign($y_j - y_k$)
2. Sum all signs → test statistic S
3. Positive S → upward trend; Negative S → downward trend

Mann-Kendall Test — The Idea

Given a time series $y_1, y_2, \ldots, y_n$, the Mann-Kendall test statistic S is defined as:

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^{n} \operatorname{sgn}(y_j - y_k)$$

where the sign function is:

$$\operatorname{sgn}(x) = \begin{cases} +1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Under the null hypothesis H_0 (no trend), S is approximately normally distributed with:

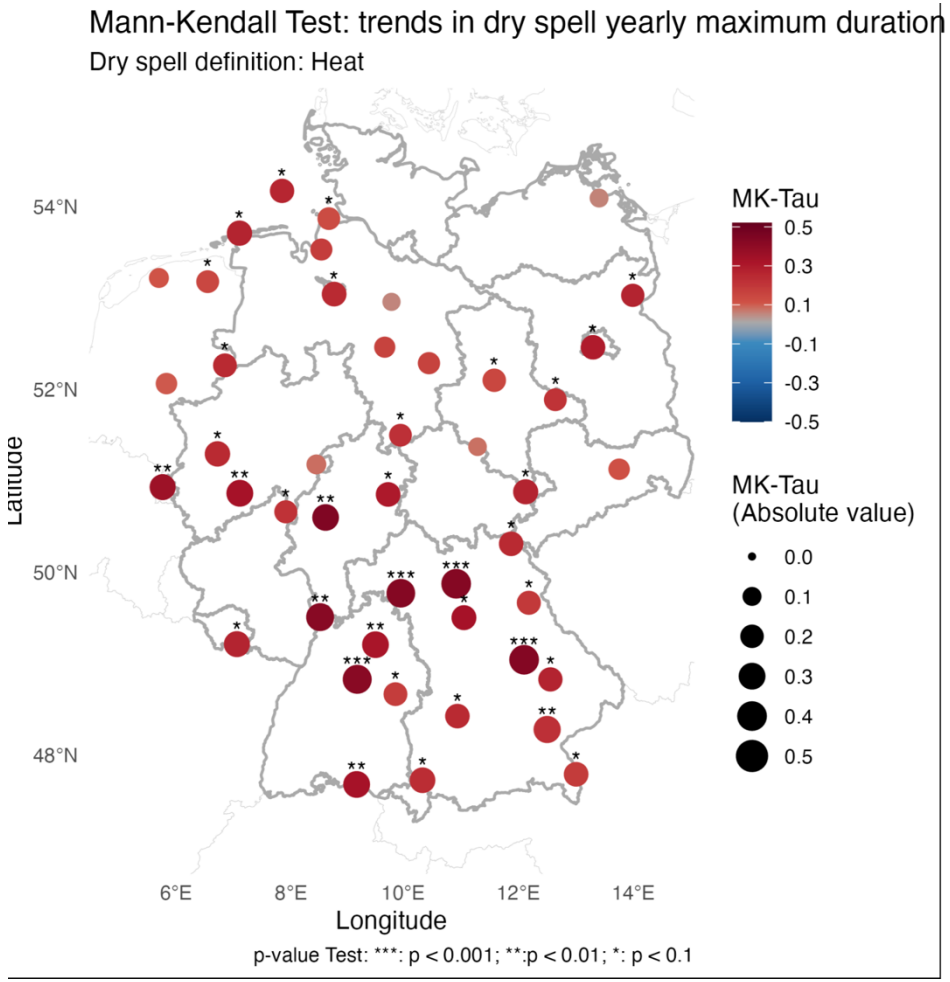
- Mean: $E[S] = 0$
- Variance: $\operatorname{Var}(S) = \frac{n(n-1)(2n+5)}{18}$

If there are tied groups, the variance is adjusted:

$$\operatorname{Var}(S) = \frac{n(n-1)(2n+5) - \sum_{p=1}^{g} t_p(t_p - 1)(2t_p + 5)}{18}$$

where g is the number of tied groups and t_p is the number of data points in the p-th tied group.

Mann-Kendall Test

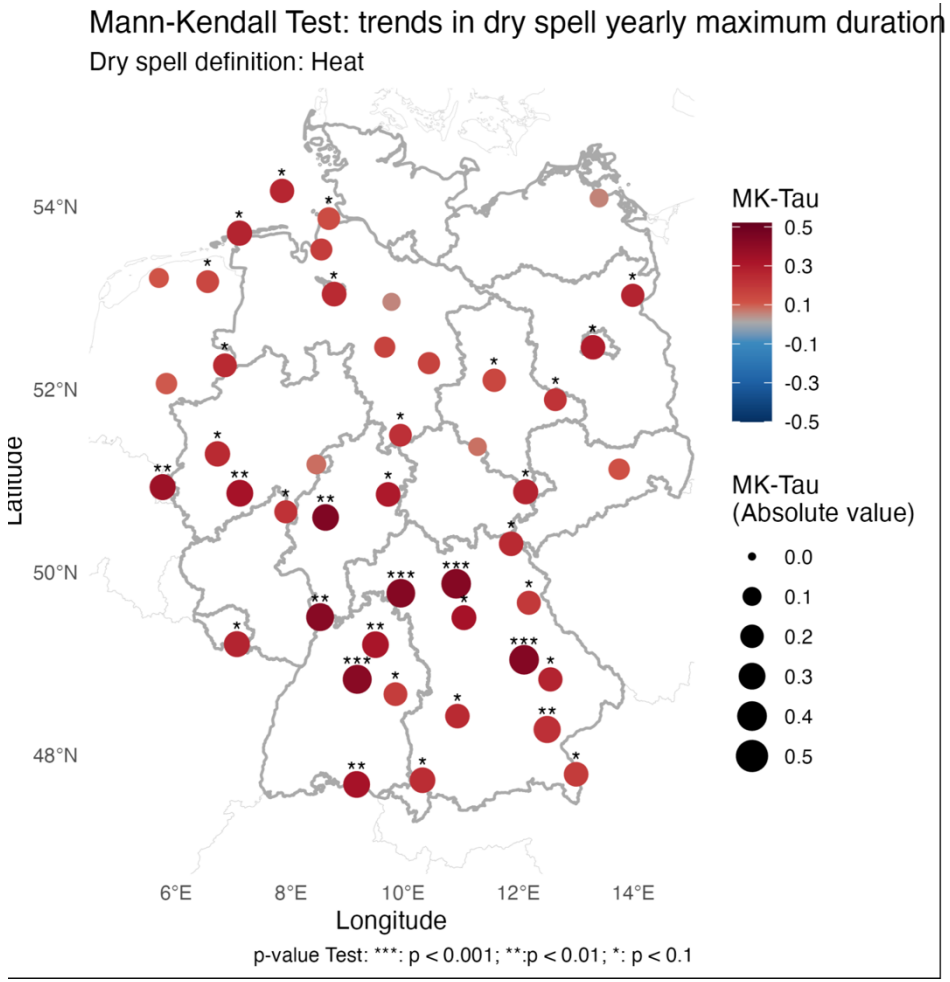


- **Decision Making**
  - Compute Z-statistic

$$Z = \begin{cases} \frac{S-1}{\sqrt{\mathrm{Var}(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S+1}{\sqrt{\mathrm{Var}(S)}} & \text{if } S < 0 \end{cases}$$

  - $|Z| > Z_{1-\alpha/2} \rightarrow$ Reject H_0 (trend exists)
  - Positive Z $\rightarrow$ upward trend
  - Negative Z $\rightarrow$ downward trend

Mann-Kendall Test

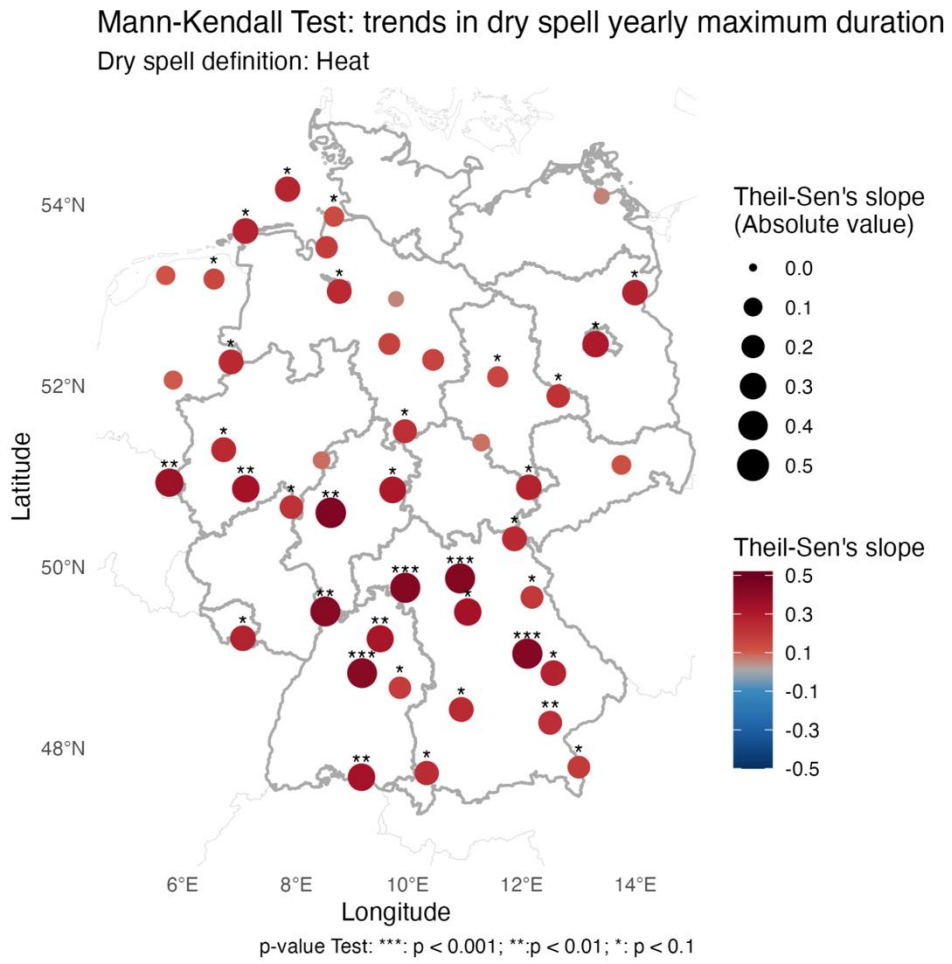


- The Mann-Kendall Tau (τ) is the normalized S:

$$\tau = \frac{S}{n(n-1)/2}$$

- Tau is limited to +/- 1, the closer to the limits, there is stronger monotonic change

Theil-Sen Slope Estimator



- While the MK-test evaluates the existence of a trend and infers its intensity, it does not provide a rate of change (slope)
- The TS-slope estimates the **slope** (rate of change) of a trend using a non-parametric approach.
- Defined as the **median** of all pairwise slopes:

$$\beta = \text{median}[(y_j - y_k) / (j - k)] \text{ for all } k < j$$

Time series statistical models

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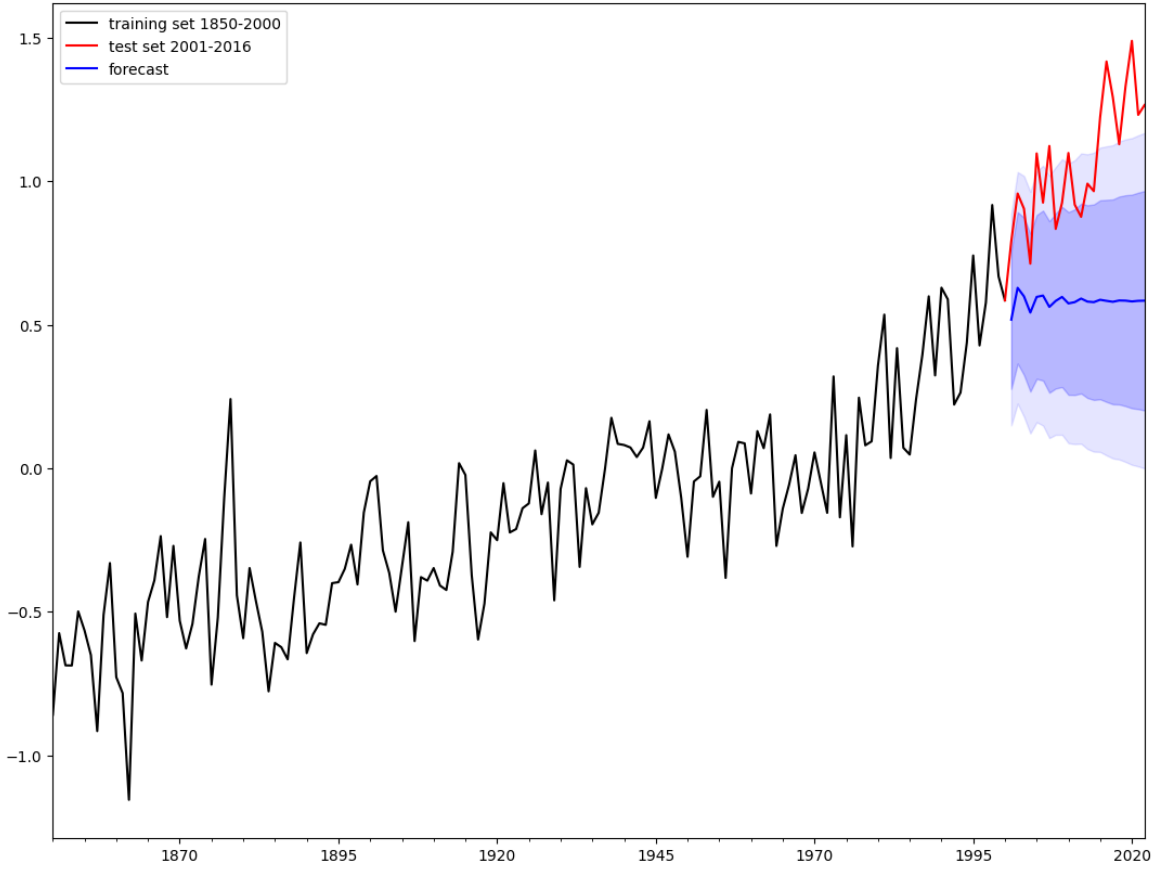
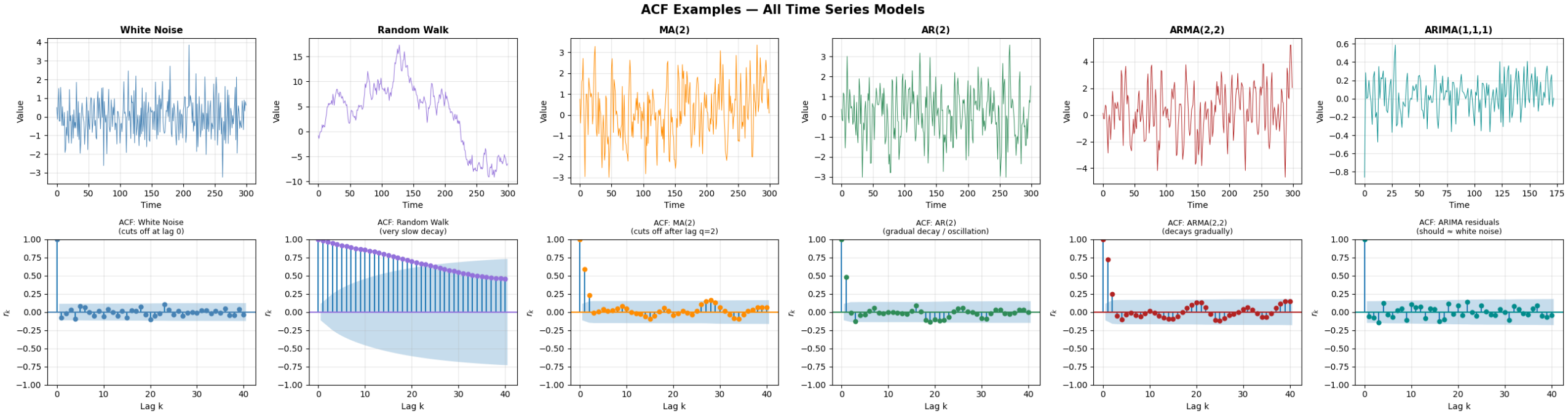
Time series statistical models



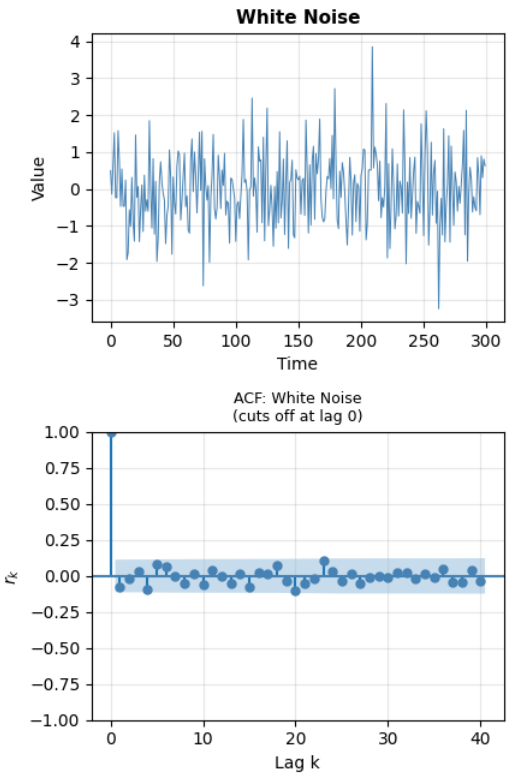
- We model time series to:
  - Understand what is driving our data (trend, seasonality, persistence, etc.)
  - Forecasting
  - Simulating (generate plausible scenarios)

Autocorrelation and ts-models

We can use the concept of autocorrelation we learned earlier to evaluate different models



Time series models – White noise



- If the random variables which make up y_t are uncorrelated, have mean 0 and variance σ^2, then y_t is stationary with an autocovariance function of

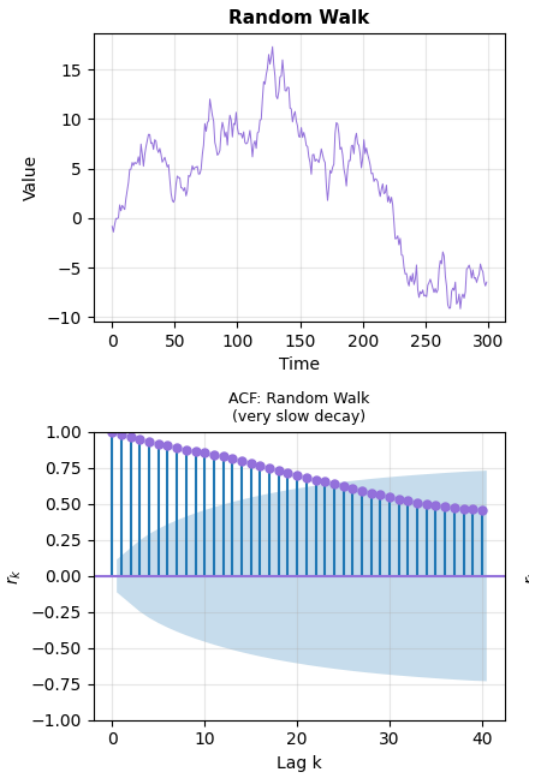
$$\gamma_w(s,t) = \mathrm{cov}(w_s, w_t) = \begin{cases} \sigma_w^2 & s = t \\ 0 & \text{otherwise} \end{cases}$$

- A particularly useful white noise series is *Gaussian white noise*, wherein the w_t are independent and identically distributed (*iid*) normal random variables, with mean 0 and variance σ^2.

$$w_t \sim \text{iid } N(0, \sigma_w^2)$$

- **Autocorrelation**: No structure whatsoever. The series has no memory — knowing y_{t-1} tells you nothing about y_t .This is the ideal residual pattern.

Time series models – Random Walk



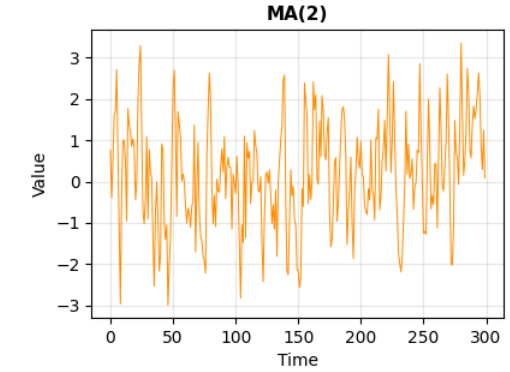
- A model for analyzing a series, which is the sum of a deterministic *trend* series and a stationary *noise* series, is the so called random walk with drift model, given by

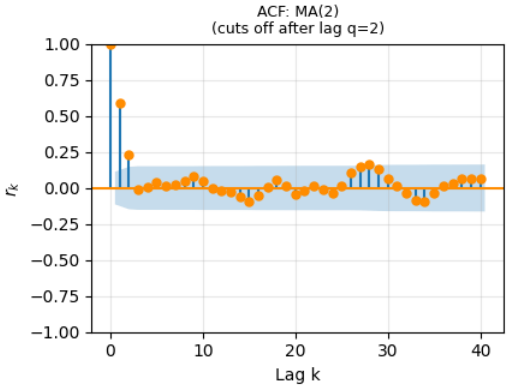
$$y_t = \delta + y_{t-1} + w_t,$$

- The constant δ is called the drift, and if δ=0, then it is simply called *random walk*.

- **Autocorrelation** : The series has an infinitely long memory. Values accumulate and never dissipate. This is the hallmark of a non-stationary, unit-root process.

Time series models – Moving average





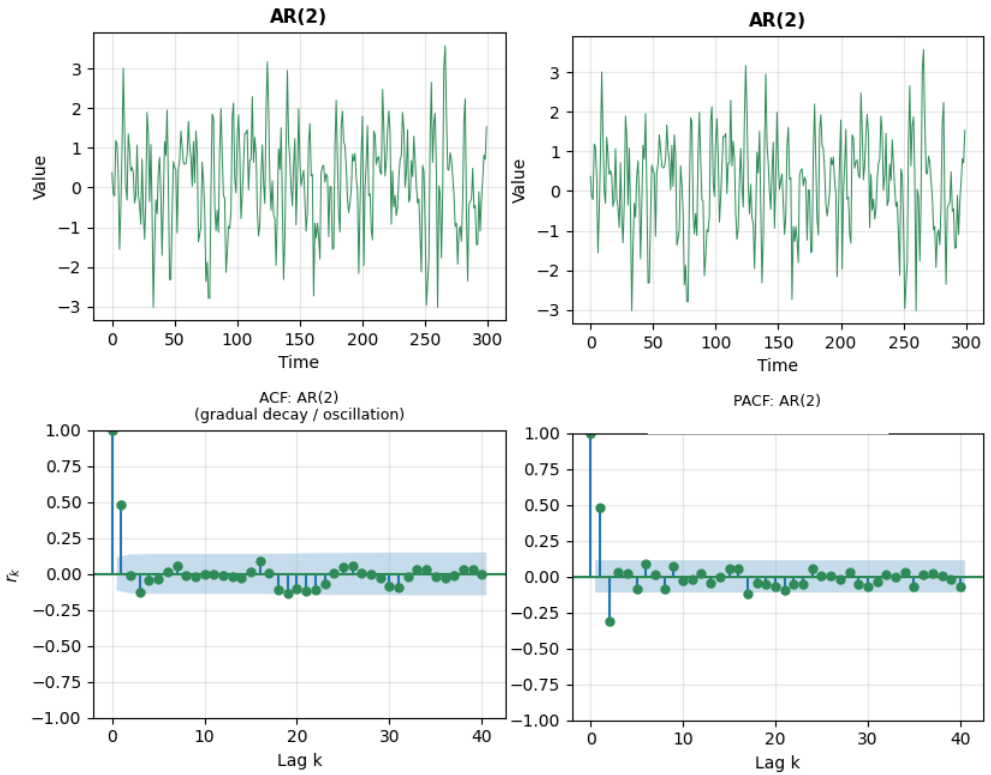
- The moving average process of order q is denoted MA(q) and defined by

$$y_t = \sum_{j=0}^{q} \theta_j w_{t-j},$$

- θ are fixed constants, with $\theta_0 = 1$. w is a variable with mean 0 and variance σ^2 (white noise)

- **Autocorrelation** : The series only “remembers” a limited number (2) of previous values with a hard cutoff beyond that

Time series models – Autoregressive



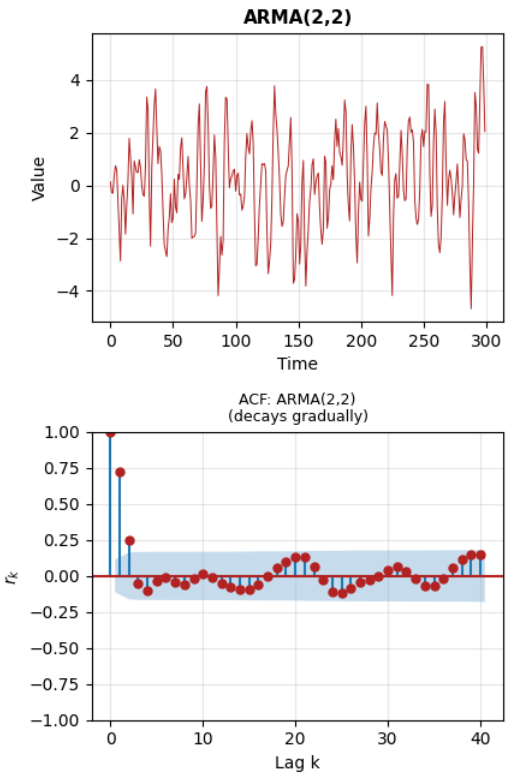
- The autoregressive process of order p is denoted AR(p), and defined

$$y_t = \sum_{j=1}^{p} \phi_j y_{t-j} + w_t,$$

- ϕ are fixed constants (note there is no 0th ϕ). w is a variable with mean 0 and variance σ^2 (white noise)

- **Autocorrelation** : Gradual decay due to memory of own past values. A hard cutoff is seen on PACF

Time series models – Autoregressive moving average (ARMA)



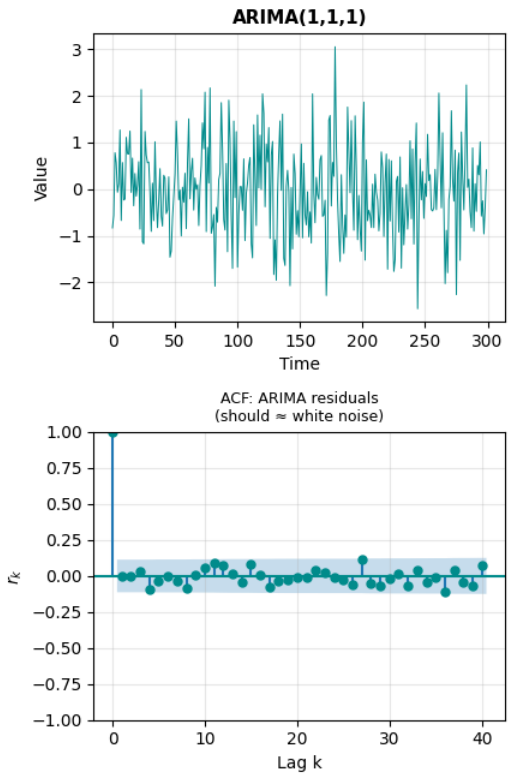
- The autoregressive moving average model (ARMA) combines the two components *autoregression* (AR) and *moving average* (MA) to model stationary processes

$$y_t = \underbrace{\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p}}_{\text{AR part}} + \underbrace{\theta_1 \omega_{t-1} + \cdots + \theta_q \omega_{t-q} + \omega_t}_{\text{MA part}}$$

- The notation ARMA(p, q) refers to the model with p autoregressive terms and q moving-average terms. This model contains the AR(p) and MA(q) models

- **Autocorrelation** : Both components (AR and MA) contribute to the autocorrelation structure. either the ACF nor the PACF show cut off

Time series models – Autoregressive integrated moving average (ARIMA)

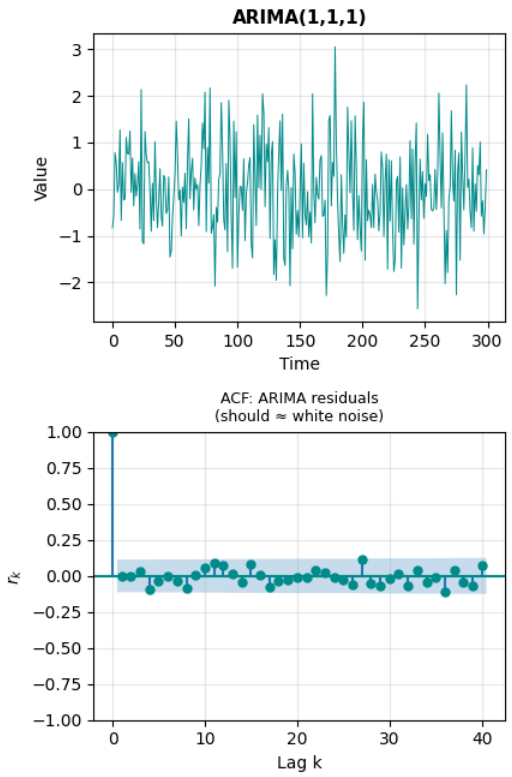


- The ARIMA model broadens the ARMA model by allowing differencing of terms, therefore being applicable to non-stationary data

$$y'_t = \phi_1 y'_{t-1} + \ldots + \phi_p y'_{t-p} + \theta_i w_{t-1} + \ldots + \theta_q w_{t-q}$$

- An ARIMA process of order p for the AR part, order q for the MA part and differences d is denoted as ARIMA(p,d,q), where
  - p− order of the autoregressive part,
  - d− degree of first differencing involved and
  - q− order of the moving average part.
- **Autocorrelation** : For ARIMA we look into the residuals and expect see white noise behaviour

Time series models – Autoregressive integrated moving average (ARIMA)



- The basic idea is that if differencing the data at some order d produces an ARMA process, then the original process is said to be ARIMA. Often, a single difference will suffice to yield a stationary series.
- Many of the models we have already discussed are special cases of the ARIMA model

White noise	ARIMA (0,0,0)
Random Walk	ARIMA (0,1,0)
Moving average	ARIMA (0,0,q)
Autoregression	ARIMA (p,0,0)
ARMA	ARIMA (p,0,q)

A little more on ARIMA and forecasting

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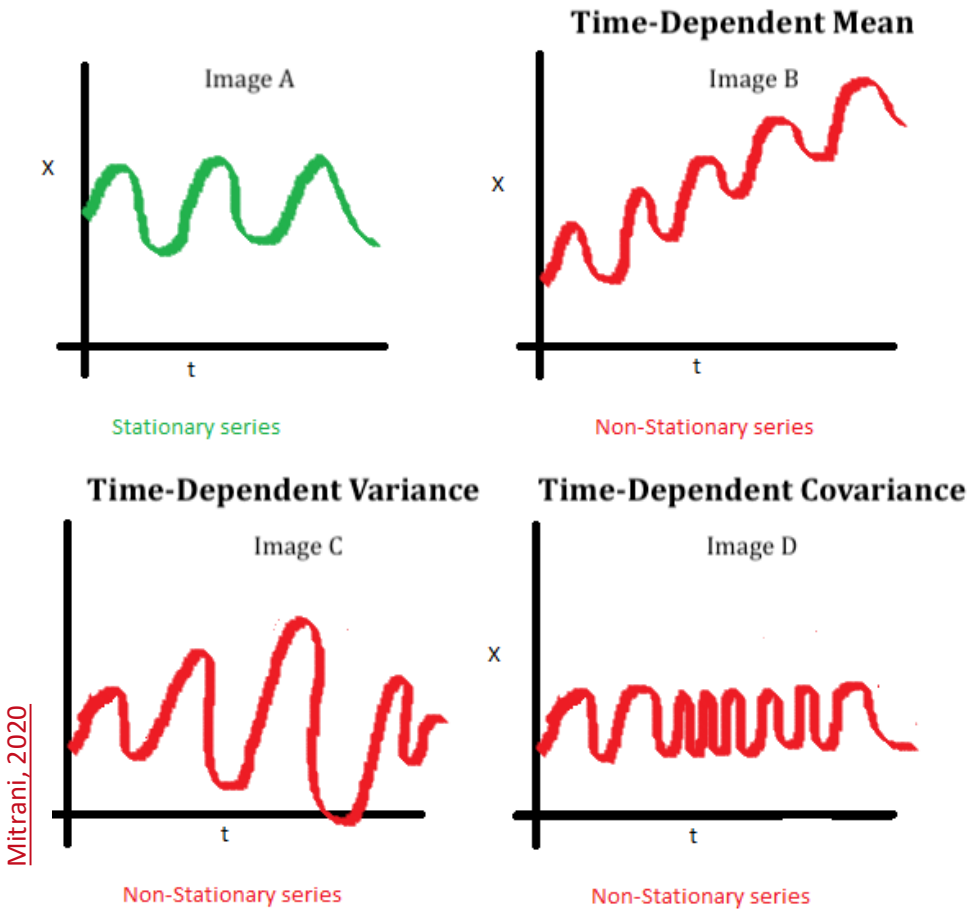
Stationarity – A Critical Assumption



Mitrani, 2020

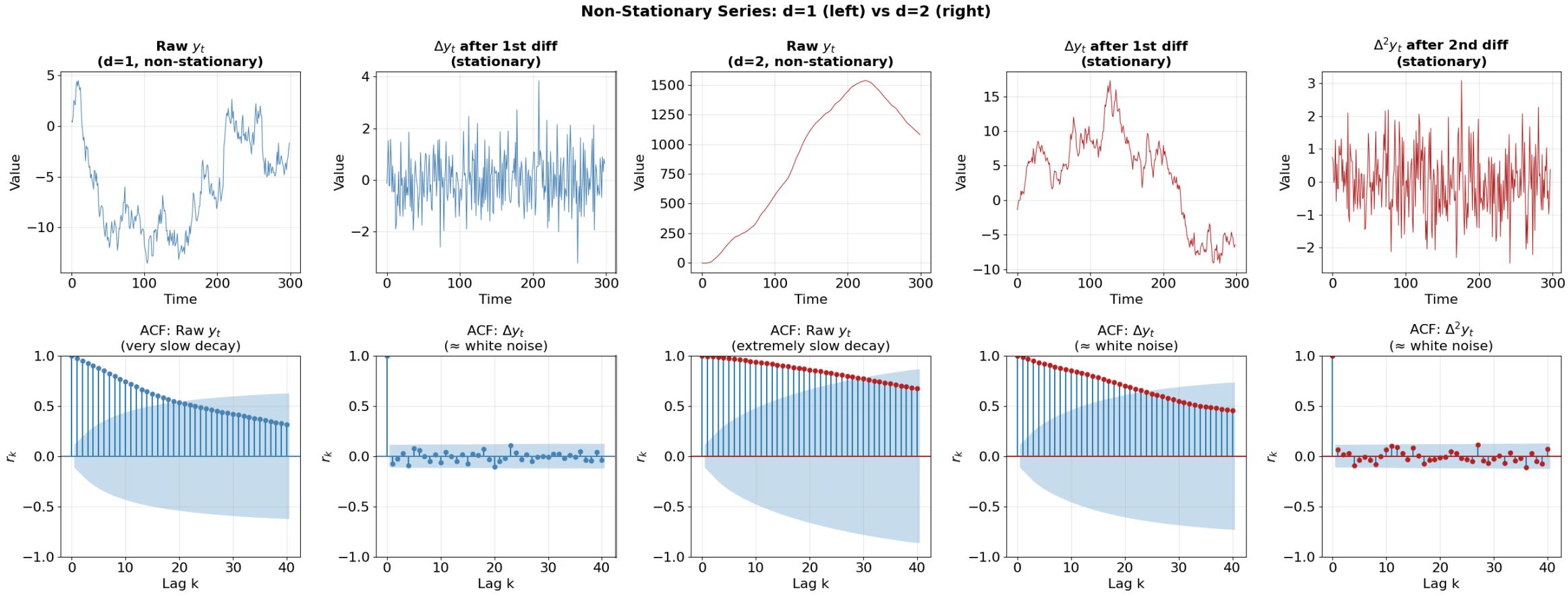
- Most time series models require **stationarity**
- A stationary time series has:
  - Constant mean over time
  - Constant variance over time
  - Autocovariance depends only on the lag, not on time
- We can evaluate a series stationarity with the KPSS test

Non-stationary series can often be made stationary by differencing

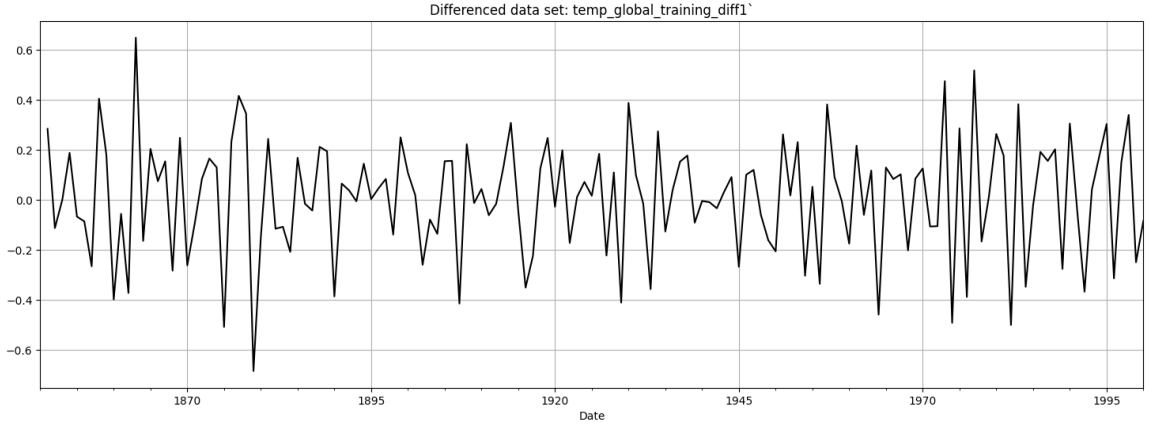
Stationarity – A Critical Assumption

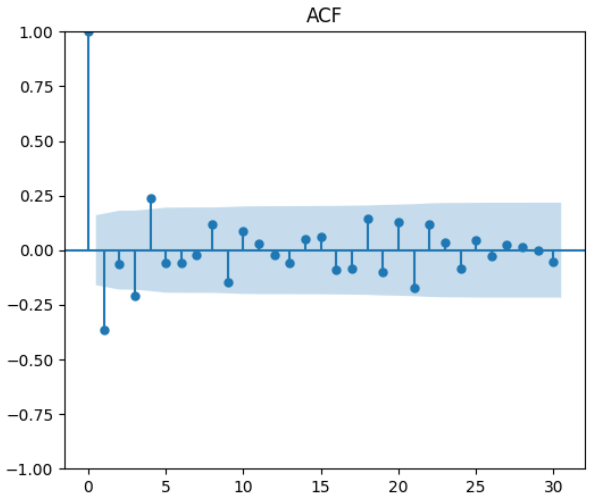
- **First difference:** $y'_t = y_t - y_{t-1}$
- **Second difference:** $y''_t = y'_t - y'_{t-1}$
- …

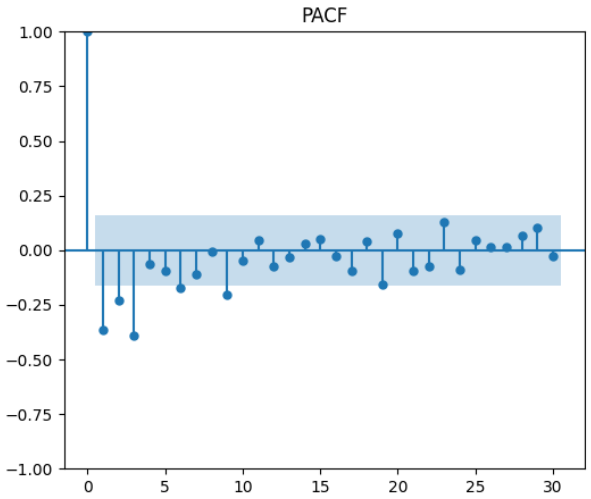
After differencing, the series should be stationary



Chooding *p* and *q*

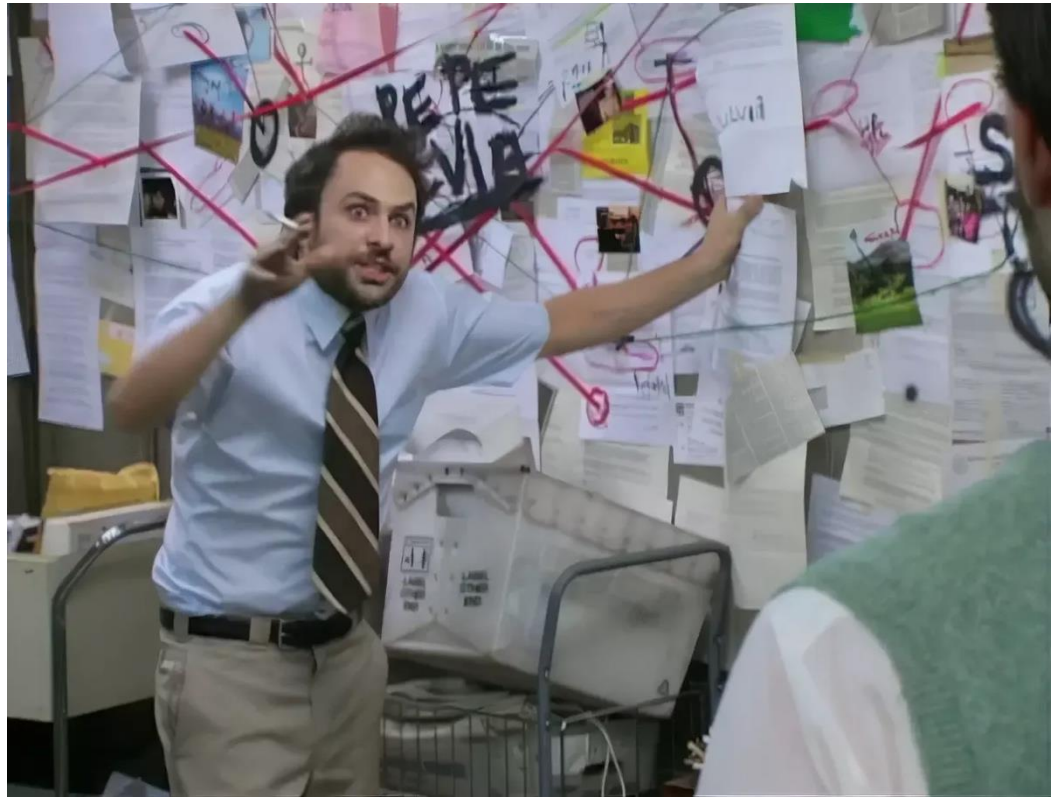






- Evaluate the shapes of ACF and PACF.
- *Generally:*
  - PACF: AR order (p)
  - ACF: MA order (q)
- **When in doubt, use information criteria (lowest is best):**
  - $\text{AIC} = -2\log L + 2k$
  - $BIC = -2\log L + k\log n$

ARIMA Modelling Workflow



1. Visualize the time series
2. Test for stationarity. Difference if necessary (determine *d*)
3. Examine ACF and PACF to identify p and q
4. Fit the ARIMA(p, d, q) model
5. Diagnose residuals (should be white noise)
6. **Forecast future values**

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Thanks!